Kaya Omer

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Education & Awards

University of Toronto — Toronto, ON
Bachelor of Science in Computer Science, Focus in Artificial Intelligence — *Sep. 2023 – May 2027*

President's Scholar of Excellence — Associated with University of Toronto
Awarded to the most highly qualified undergraduate students. — *March 2023*

Janet Faulds and M. Ann Marshall Family Award. — St. Michael's College at University of Toronto
Awarded to the qualified students at St Michael's College. — *Nov 2023*

Experience

AI Researcher Intern — May 2026 – Present
The Matter Lab — *Toronto, ON*

- Engineering specialized AI agents for **quantum chemistry** workflows using **PySCF** and **ORCA** to automate electronic structure calculations, molecular property computation, and API interfacing.
- Architecting autonomous LLM-driven scientific discovery pipelines ("**El Agente**"), building agentic workflows that orchestrate quantum chemistry tools, external APIs, and data retrieval systems.
- Developing an AI-native framework integrating LLM-based agents with internal web interfaces, enabling automated execution of complex computational chemistry protocols.

Software Engineer Intern — Jan 2026 – April 2026
CIBC — *Toronto, ON*

- Engineered internal AI integrations to automate workflows; built data pipelines using Python, Apache Spark, and Databricks to process large-scale financial data.
- Implemented MLOps workflows using Azure AI Studio, Azure ML, and Azure DevOps, including CI/CD pipelines for automated model deployment.

Machine Learning Research Assistant — April 2026 – Present
CECCS — *Toronto, ON*

- Refactoring a Python-based SDG classification pipeline, applying NLP and LLM-based tools to resolve data quality issues and improve precision of sustainability course classification at scale.

Research Assistant / Software Engineer — Sep 2025 – Dec 2025
Acceleration Consortium — *Toronto, ON*

- Engineered a full Computer Vision pipeline for a Self-Driving Lab; implemented Structure from Motion (SfM) using OpenCV for 3D reconstruction and dimensional accuracy analysis with NumPy.
- Developed a Robotic Data Acquisition System integrating a PiCam and stepper motor for automated 360° object scanning within a closed-loop autonomous experimentation framework.

Projects

Quantum Chemistry Agent Pipeline | *Python, PySCF, ORCA, LLMs* — May 2026

- Built LLM-driven agentic workflows to automate quantum-chemistry calculations via PySCF and ORCA, orchestrating multi-step electronic structure computations with tool-use and automated error handling.

3D Reconstruction & Automation Pipeline for SDL | *Python, OpenCV, NumPy, RPi* — Sep 2025

- Engineered a CV pipeline for a Self-Driving Lab combining SfM-based 3D reconstruction with automated robotic scanning for closed-loop material quality assessment.

KnowBody | *JavaScript, Cloudflare Workers, React, Workers AI (Llama 3.3)* — Jan 2026

- Built a research-based health platform using React and ES6+ JavaScript with real-time ML-driven grading and state management across complex daily metric inputs.

Technical Skills

Languages: Python, C/C++, Java, SQL, Swift, JavaScript, HTML/CSS, R, Assembly
Frameworks: PyTorch, TensorFlow, React, Flask, FastAPI
Developer Tools: Git, Docker, CUDA, GCP, Azure AI Studio, Databricks, Linux
Libraries: PySCF, SciPy, OpenCV, pandas, NumPy, Matplotlib, ORCA
AI/ML: LLM Agents, Agentic Workflows, Quantum Chemistry Automation, MLOps, Computer Vision